

1. Conceptual Data Model

- Entities:
- Customer
- Represents individual customers loaded from a source CSV
- Order
- Represents individual orders loaded from a source CSV
- OrderSummary
- Represents historical and current joined records of Customer and Order with SCD Type 2 logic
- CustomerAggregateSpend
- Represents daily aggregate spend per customer
- Relationships:
- Customer 1 -- * Order (via CustId)
- Customer * -- * OrderSummary (via CustId, SCD Type 2 history maintained)
- Order * -- * OrderSummary (via CustId)
- OrderSummary * -- * CustomerAggregateSpend (via Name, Date)
- Business Definitions:
- Customer: Entity containing customer profile data (CustId, Name, EmailId, Region)
- Order: Entity containing purchase records (OrderId, ItemName, PricePerUnit, Qty, Date, CustId)
- OrderSummary: Entity representing SCD Type 2 historical join of Customer and Order (including all fields from both)
- CustomerAggregateSpend: Entity representing aggregated spend per Name/Date

2. Logical Data Model

- Entities and Attributes:
- Customer
- CustId (Primary Key)
- Name
- EmailId
- Region
- Order
- OrderId (Primary Key)
- ItemName
- PricePerUnit
- Qty

- Date
- CustId (Foreign Key to Customer)
- OrderSummary
- CustId
- Name
- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- [SCD Type 2 fields: StartDate, EndDate, Active] (per business rules, not explicitly defined)
- CustomerAggregateSpend
- Name
- TotalAmount
- Date
- Relationship and Cardinality Rules:
- Customer to Order: 1-to-many (1 customer can have multiple orders)
- Order and Customer to OrderSummary: Many-to-many (history-tracked relationship), joined on CustId
- OrderSummary to CustomerAggregateSpend: Many-to-many, grouped by Name/Date
- Keys:
- Each entity's primary keys as described
- Foreign key: Order.CustId references Customer.CustId
- Normalization Notes:
- Entities normalized to 3NF
- No repeating groups
- Non-key attributes depend only on primary key
- No transitive dependencies
- Business Rules per Entity:
- Order: TotalAmount = PricePerUnit × Qty (derived, stored in downstream tables)
- Customer/Order: Remove records with "Null"/Null literals, remove duplicate records

- OrderSummary: Apply SCD Type 2 change handling—maintain historical records with status indication

3. Physical Data Model

- CustId	VARCHAR	PK
- Name	VARCHAR	
- EmailId	VARCHAR	
- Region	VARCHAR	

- Table: customer

- OrderId	VARCHAR	PK
- ItemName	VARCHAR	
- PricePerUnit	DECIMAL	
- Qty	INTEGER	
- Date	DATE	
- CustId	VARCHAR	FK to customer.CustId

- Table: order

- CustId	VARCHAR	
- Name	VARCHAR	
- EmailId	VARCHAR	
- Region	VARCHAR	
- OrderId	VARCHAR	
- ItemName	VARCHAR	
- PricePerUnit	DECIMAL	
- Qty	INTEGER	
- Date	DATE	
- StartDate	DATE	SCD2 tracking (if implemented)
- EndDate	DATE	SCD2 tracking (if implemented)
- Active	BOOLEAN	SCD2 tracking (if implemented)

- Table: ordersummary

- Name	VARCHAR
- TotalAmount	DECIMAL
- Date	DATE

- Table: customeraggregatespend
- Indexing Strategy:
 - customer: PK on CustId
 - order: PK on OrderId, index on CustId
 - ordersummary: indexes on CustId, OrderId; if SCD2, composite index on (CustId, StartDate)
 - customeraggregatespend: index on Name, Date
- Partitioning, Clustering, Storage:
 - ordersummary, customeraggregatespend partitioned by Date if volume is high
 - Tables stored as Delta tables (per requirements)
 - No explicit clustering or Z-ordering defined in requirements
- Constraints:
 - PK and FK as noted
 - Unique constraint on customer.CustId, order.OrderId
 - Referential integrity between order.CustId and customer.CustId
- Performance Optimization:
 - Remove null or duplicate records before load
 - Precompute aggregates for customeraggregatespend to optimize query performance

4. Naming Conventions

- All table and column names in lower_case, snake_case as per provided requirements
- Primary keys: _id where possible (e.g., cust_id, order_id)
- Foreign keys use identical names to referenced keys
- Computed columns have descriptive names (e.g., TotalAmount)

5. Data Dictionary

Entity	Attribute	Definition	Type	Nullable	Notes
customer	CustId	Customer unique identifier	VARCHAR	No	PK, external source
customer	Name	Customer name	VARCHAR	No	
customer	EmailId	Email address of customer	VARCHAR	Yes	
customer	Region	Geographic region	VARCHAR	Yes	
order	OrderId	Order unique identifier	VARCHAR	No	PK, external source
order	ItemName	Name of purchased item	VARCHAR	No	
order	PricePerUnit	Unit price of item	DECIMAL	No	
order	Qty	Quantity ordered	INTEGER	No	

order	Date	Order date	DATE	No	
order	CustId	Customer unique identifier	VARCHAR	No	FK to customer.CustId
ordersummary	CustId	Customer unique identifier	VARCHAR	No	
ordersummary	Name	Customer name	VARCHAR	No	
ordersummary	EmailId	Email address of customer	VARCHAR	Yes	
ordersummary	Region	Geographic region	VARCHAR	Yes	
ordersummary	OrderId	Order unique identifier	VARCHAR	No	
ordersummary	ItemName	Name of purchased item	VARCHAR	No	
ordersummary	PricePerUnit	Unit price of item	DECIMAL	No	
ordersummary	Qty	Quantity ordered	INTEGER	No	
ordersummary	Date	Order date	DATE	No	
ordersummary	StartDate	SCD2 record start date	DATE	Yes	SCD Type 2 (if implemented)
ordersummary	EndDate	SCD2 record end date	DATE	Yes	SCD Type 2 (if implemented)
ordersummary	Active	Record active status for SCD2	BOOLEAN	Yes	SCD Type 2 (if implemented)
customeraggregatespend	Name	Customer name	VARCHAR	No	
customeraggregatespend	TotalAmount	Aggregate spend per day	DECIMAL	No	sum(PricePerUnit × Qty)
customeraggregatespend	Date	Aggregate date	DATE	No	

6. ERD Diagram

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